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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

2 - Proofs in Propositional Logic - Solutions

2.1) Match each of the following tautologies to one of the logical arguments.

Tautologies

(1) $p \vee \neg p$
(2) $p \wedge q \rightarrow p$
(3) $p \rightarrow p \vee q$
(4) $(p \rightarrow q) \rightarrow (\neg q \rightarrow \neg p)$
(5) $((p \rightarrow q) \wedge (\neg p \rightarrow q)) \rightarrow q$
(6) $(((p \vee q) \rightarrow r) \wedge p) \rightarrow r$

Arguments

(a) If x is a real number such that $0 \leq x \leq 1$, then $x \leq 1$.
(b) Let n be a natural number such that n is divisible by 3. Then n is divisible by 2 or n is divisible by 3.
(c) John gets frequency if he does his work in C or Java. John did the work in C. So John obtained frequency.
(d) Let x be a real number; we have $x < 0$ or $x \geq 0$.
(e) Let n be an integer; if n is divisible by 4 then n is divisible by 2. Then, if n is not divisible by 2, then is not divisible by 4.
(f) Let n be an integer; if n is even, then $n^2 - n = (n-1)n$ is even; if n is odd, then $n^2 - n = (n-1)n$ is even. Therefore $n^2 - n$ is always even.

Solution:

Tautologies	Arguments
(1)	(d)
(2)	(a)
(3)	(b)
(4)	(e)
(5)	(f)
(6)	(c)

2.2) Indicate whether the following deductions are valid. If so, indicate the rules used in each step. If not valid, indicate which step it is not correct and if there is a correct deduction with the same premises and conclusion.

a)

1	$p \vee q$
2	| p
3	| p
4	| q
5	| q
6	$p \wedge q$

b)

1	$(p \wedge q) \vee r$
2	| $p \wedge q$
3	| p
4	| $p \vee r$
5	| r
6	| $p \vee r$
7	$p \wedge r$

c)

1	$\neg(p \vee \neg q)$
2	| $\neg q$
3	| $p \vee \neg q$
4	| $\mathbf{F}$
5	$\neg\neg q$
6	q

d)

1	$p \to q$
2	| p
3	| q
4	| $\neg p \vee q$
5	$\neg p \vee q$

e)

1	$p \to \neg p$
2	| p
3	| $\neg p$
4	| $\mathbf{F}$
5	$\neg p$

f)

1	$\neg p \to \neg q$
2	| q
3	| $\neg q$
4	| $\mathbf{F}$
5	| p
6	$q \to p$

Solution:

(a) The proof is not valid. We cannot justify step 6, since we cannot apply $\wedge$I on premises that appear in sub-deductions that already ended. Furthermore, it is not possible to build such proof, since $p \vee q \not\vdash p \wedge q$. Let $\nu(p) = \mathbf{T}$ and $\nu(q) = \mathbf{F}$, then $\models_\nu p \vee q$ but $\not\models_\nu p \wedge q$.

(b) The proof is not valid. We cannot justify step 7, since by $\vee$E one would conclude $p \vee r$, not $p \wedge r$. Once again, it is not possible to build such proof, since $(p \wedge q) \vee r \not\vdash p \wedge r$. Let $\nu(p) = \mathbf{F}$ and $\nu(r) = \mathbf{T}$, then $\models_\nu (p \wedge q) \vee r$ but $\not\models_\nu p \wedge r$.

(c) The proof is valid:

1	$\neg(p \vee \neg q)$	
2	| $\neg q$	
3	| $p \vee \neg q$	$\vee$I, 2
4	| $\mathbf{F}$	$\mathbf{F}$I, 1, 3
5	$\neg\neg q$	$\neg$I, 2–4
6	q	$\neg$E, 5

(d) The proof is not valid. It is not possible to justify step 5 by any rule of the system. From the sub-deduction one could conclude $p \to (\neg p \vee q)$, but not $\neg p \vee q$. However the logic consequence is valid and one can build a correct deduction (in fact, more than one).

1	$p \to q$	
2	$\quad \neg(\neg p \vee q)$	
3	$\quad\quad p$	
4	$\quad\quad q$	$\to$E, 1, 3
5	$\quad\quad \neg p \vee q$	$\vee$I, 4
6	$\quad\quad \mathbf{F}$	**F**I, 2, 5
7	$\quad \neg p$	$\neg$I, 3–6
8	$\quad \neg p \vee q$	$\vee$I, 7
9	$\quad \mathbf{F}$	**F**I, 2, 8
10	$\neg\neg(\neg p \vee q)$	$\neg$I, 2–9
11	$\neg p \vee q$	$\neg$E, 10

1	$p \to q$	
2	$p \vee \neg p$	TE
3	$\quad p$	
4	$\quad q$	$\to$E, 1, 3
5	$\quad \neg p \vee q$	$\vee$I, 4
6	$\quad \neg p$	
7	$\quad \neg p \vee q$	$\vee$I, 6
8	$\neg p \vee q$	$\vee$E, 2, 3–5, 6–7

Let us look at the proof on the left step by step:

$p \to q$
$\ldots$
$\neg p \vee q$

Since we want to deduce a disjuntion, one might feel inclined to derive $\neg p$ or q, but note however that neither $\neg p$ nor q are a semantic consequence of $p \to q$, therefore such deductions will not hold. And since the only premise we have is an implication, then we need additional premises. Since our conclusion is not an implication, the only rule (without considering derived premises) that allows us to assume extra premises is the introduction of $\neg$, therefore our proof follows by contradiction:

$p \to q$	
$\quad \neg(\neg p \vee q)$	
$\quad\quad \ldots$	
$\quad\quad \mathbf{F}$	
$\neg\neg(\neg p \vee q)$	$\neg$I
$\neg p \vee q$	$\neg$E

Now note that the added premise is a negation, which can only be used to introduce **F**. Therefore one needs to deduce $\neg p \vee q$, which can be deduced from either $\neg p$ or q. And note that if we add

the hypothesis p, we can then use it with $p \to q$ to deduce q, which introduces a contradiction.

1	$p \to q$	
2	$\quad \neg(\neg p \vee q)$	
3	$\quad\quad p$	
4	$\quad\quad q$	$\to$E, 1, 3
5	$\quad\quad \neg p \vee q$	$\vee$I, 4
6	$\quad\quad \mathbf{F}$	**F**I, 2, 5
7	$\quad \ldots$	
8	$\quad \mathbf{F}$	**F**I, 2, 8
9	$\neg\neg(\neg p \vee q)$	$\neg$I, 2–9
10	$\neg p \vee q$	$\neg$E, 10

Now we can use the negation introduction to get $\neg p$, then getting another contradiction and completing the deduction:

1	$p \to q$	
2	$\quad \neg(\neg p \vee q)$	
3	$\quad\quad p$	
4	$\quad\quad q$	$\to$E, 1, 3
5	$\quad\quad \neg p \vee q$	$\vee$I, 4
6	$\quad\quad \mathbf{F}$	**F**I, 2, 5
7	$\quad \neg p$	$\neg$I, 3–6
8	$\quad \neg p \vee q$	$\vee$I, 7
9	$\quad \mathbf{F}$	**F**I, 2, 8
10	$\neg\neg(\neg p \vee q)$	$\neg$I, 2–9
11	$\neg p \vee q$	$\neg$E, 10

(e) The proof is valid:

1	$p \to \neg p$	
2	$\quad p$	
3	$\quad\quad \neg p$	$\to$E, 1, 2
4	$\quad\quad \mathbf{F}$	**F**I, 2, 3
5	$\neg p$	$\neg$I, 2–4

(f) The proof is not valid since there is no rule that justifies step 3. However the logic consequence

is valid and one can build a correct deduction.

1	$\neg p \to \neg q$	
2	$\quad q$	
3	$\quad \neg\neg q$	$\neg\neg$I, 2
4	$\quad \neg\neg p$	MT, 1, 3
5	$\quad p$	$\neg$E, 4
6	$q \to p$	$\to$I, 2–5

And without using derived rules:

1	$\neg p \to \neg q$	
2	$\quad q$	
3	$\quad\quad \neg p$	
4	$\quad\quad \neg q$	$\to$E, 1, 3
5	$\quad\quad \mathbf{F}$	**F**I, 2, 4
6	$\quad \neg\neg p$	$\neg$I, 3–5
7	$\quad p$	$\neg$E, 6
8	$q \to p$	$\to$I, 2–7

2.3) Using rules of logical inference, build a proof for the conclusion "It rained" from the hypothesis "If it doesn't rain or if there is no fog, then the sailing will take place and the rescue demonstration will be carried out", "If the sailing competition takes place then the trophy will be awarded" and "The trophy was not awarded".

Solution: Let us first associate propositional variables to each atomic sentence.

- p: "It rains"
- q: "There is fog"
- r: "The sailing takes place"
- s: "Thee rescue demonstration is carried out"
- t: "The trophy is awarded"

Then one wants to deduce:

$$(\neg p \vee \neg q) \to (r \wedge s), r \to t, \neg t \vdash p$$

Let us once more look at this deduction step by step.

$(\neg p \vee \neg q) \to (r \wedge s)$
$r \to t$
$\neg t$
$\ldots$
p

Given that we have $r \to t$ and $\neg t$ then one might feel inclined to use modus tollens. However this would allow us to deduce $\neg r$, from which there is nothing that we can directly deduce from the premises. On the other hand, if we decide to do a proof by contradiction, then assuming $\neg p$ would allow us to build

$\neg p \vee \neg q$, which would allow us to use the first premise to obtain $r \wedge s$, from which we can then get both r and s.

$(\neg p \vee \neg q) \to (r \wedge s)$	
$r \to t$	
$\neg t$	
$\neg p$	
$\neg p \vee q$	$\vee$I
$r \wedge s$	$\to$E
$\ldots$	
F	
$\neg\neg p$	$\neg$I
p	$\neg$E

But if we get r then using the second premise will allow us to get t, which can then be used with the last premise to deduce **F** thus completing the deduction.

1	$(\neg p \vee \neg q) \to (r \wedge s)$	
2	$r \to t$	
3	$\neg t$	
4	$\neg p$	
5	$\neg p \vee \neg q$	$\vee$I, 4
6	$r \wedge s$	$\to$E, 1, 5
7	r	$\wedge$E, 6
8	t	$\to$E, 2, 7
9	**F**	**F**I, 3, 8
10	$\neg\neg p$	$\neg$I, 4–9
11	p	$\neg$E, 10

2.4) Show that:

(a) $(p \wedge q) \to r, p \vdash q \to r$
Solution:

1	$(p \wedge q) \to r$	
2	p	
3	q	
4	$p \wedge q$	$\wedge$I, 2, 3
5	r	$\to$E, 1, 4
6	$q \to r$	$\to$I, 3–5

(b) $q \to p, q \to \neg p \vdash \neg q$

Solution:

1	$q \to p$	
2	$q \to \neg p$	
3	$\quad q$	
4	$\quad p$	$\to$E, 1, 3
5	$\quad \neg p$	$\to$E, 2, 3
6	$\quad \mathbf{F}$	**F**I, 4, 5
7	$\neg q$	$\neg$I, 3–6

(c) $(\neg p \vee \neg q) \vee r, q \vee r, p \vdash r$

Solution: Let us take this one step by step.

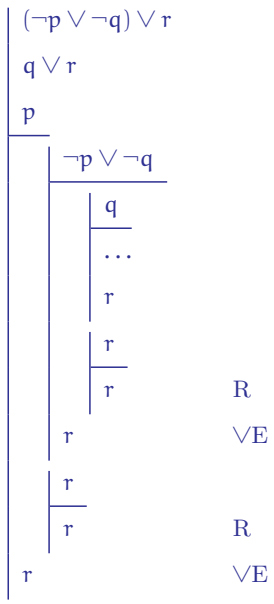
$$
\begin{array}{|l}
(\neg p \vee \neg q) \vee r \\
q \vee r \\
p \\
\hline
\quad \ldots \\
\quad r
\end{array}
$$

First notice that two of our premises are disjuntions, which we can only be used in eliminations of disjuntions. Let us start with the longest one:

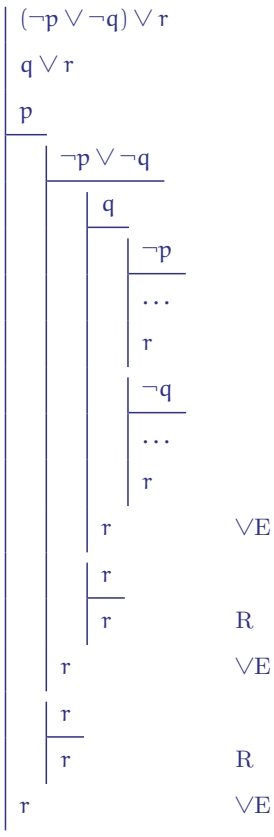
$$
\begin{array}{|lll}
(\neg p \vee \neg q) \vee r & & \\
q \vee r & & \\
p & & \\
\hline
\quad \neg p \vee \neg q & & \\
\quad\quad \ldots & & \\
\quad\quad r & & \\
\quad r & & \\
\quad\quad r & & \\
\quad r & & R \\
r & & \vee E
\end{array}
$$

Note that the second part of the disjunction trivially deduces r by repetition. Now, looking at the first deduction we have again two disjuntions that we can use to deduce r. We will start with

$q \vee r$, since one of its branches also trivially holds r:



And finally we need to use the last disjuntion:



Now notice that in the first incomplete sub-deduction we have p, q and $\neg p$ has premises and in the second p, q and $\neg q$. And in both cases we want to deduce r, which does not occur in these

premises, therefore we need to obtain r by an elimination of **F**, which can easily be done in both cases, then completing the deduction:

1	$(\neg p \vee \neg q) \vee r$	
2	$q \vee r$	
3	p	
4	$\quad \neg p \vee \neg q$	
5	$\quad\quad q$	
6	$\quad\quad\quad \neg p$	
7	$\quad\quad\quad \mathbf{F}$	**F**I, 3, 6
8	$\quad\quad\quad r$	**F**E, 7
9	$\quad\quad\quad \neg q$	
10	$\quad\quad\quad \mathbf{F}$	**F**I, 5, 9
11	$\quad\quad\quad r$	**F**E, 10
12	$\quad\quad r$	$\vee$E, 4, 6–8, 9–11
13	$\quad\quad r$	
14	$\quad\quad r$	R, 13
15	$\quad r$	$\vee$E, 2, 5–12, 13–14
16	$\quad r$	
17	$\quad r$	R, 16
18	r	$\vee$E, 1, 4–15, 16–17

(d) $p \vee \neg q, \neg r \to q, r \to \neg s, s \vdash p$

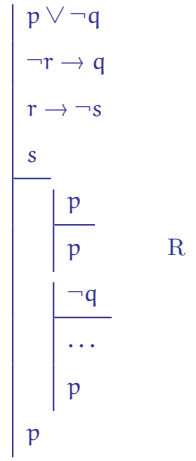
Solution:

Let us build a deduction (without derived rules) step-by-step:

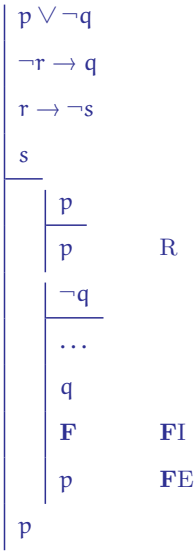
$p \vee \neg q$	
$\neg r \to q$	
$r \to \neg s$	
s	
$\ldots$	
p	

Given that one of the premises is a disjuntion where one on the branches is exactly what we want

to deduce, let us start by using that premise in an elimination:



Now notice that p does not occur in the premises that we have left on the second branch. A way to deduce p is by elimination of $\mathbf{F}$, which we could obtain if we could deduce (and we can) q. This deduction would look like:



Now, to obtain q from the premises we need $\neg r$, which leads to a proof by contradiction:

$p \vee \neg q$
$\neg r \to q$
$r \to \neg s$
s
| p
| p R
| $\neg q$
| | r
| | $\ldots$
| | $\mathbf{F}$
| $\neg r$ $\neg$I
| q $\to$E
| $\mathbf{F}$ $\mathbf{F}$I
| p $\mathbf{F}$E
p

And from r, using the third premise, we obtain $\neg s$, which combine with the last premise leads to $\mathbf{F}$, thus concluding the proof:

1	$p \vee \neg q$	
2	$\neg r \to q$	
3	$r \to \neg s$	
4	s	
5	| p	
6	| p	R, 5
7	| $\neg q$	
8	| | r	
9	| | $\neg s$	$\to$E, 3, 8
10	| | $\mathbf{F}$	$\mathbf{F}$I, 4, 9
11	| $\neg r$	$\neg$I, 8–10
12	| q	$\to$E, 2, 11
13	| $\mathbf{F}$	$\mathbf{F}$I, 7, 12
14	| p	$\mathbf{F}$E, 13
15	p	$\vee$E, 1, 5–6, 7–14

Another solution using derived rules:

1	$p \vee \neg q$	
2	$\neg r \to q$	
3	$r \to \neg s$	
4	s	
5	$\quad p$	
6	$\quad p$	R, 5
7	$\quad \neg q$	
8	$\quad \neg\neg r$	MT, 2, 7
9	$\quad r$	$\neg\neg$I, 8
10	$\quad \neg s$	$\to$E, 3, 9
11	$\quad \mathbf{F}$	**F**I, 4, 10
12	$\quad p$	**F**E, 11
13	p	$\vee$E, 1, 5–6, 7–12

And another one, a bit longer:

1	$p \vee \neg q$	
2	$\neg r \to q$	
3	$r \to \neg s$	
4	s	
5	$\quad p$	
6	$\quad p$	R, 5
7	$\quad \neg q$	
8	$\quad r \vee \neg r$	TE
9	$\quad\quad r$	
10	$\quad\quad \neg s$	$\to$E, 3, 9
11	$\quad\quad \mathbf{F}$	**F**I, 4, 10
12	$\quad\quad \neg r$	
13	$\quad\quad q$	$\to$E, 2, 12
14	$\quad\quad \mathbf{F}$	**F**I, 7, 13
15	$\quad \mathbf{F}$	$\vee$E, 8, 9–15, 12–14
16	$\quad p$	**F**E, 15
17	p	$\vee$E, 1, 5–6, 7–16

(e) $p \to (q \vee r), q \to s, r \to t \vdash p \to (s \vee t)$.

Solution:

1	$p \to (q \vee r)$	
2	$q \to s$	
3	$r \to t$	
4	$\quad p$	
5	$\quad q \vee r$	$\to$E, 1, 4
6	$\quad\quad q$	
7	$\quad\quad s$	$\to$E, 2, 6
8	$\quad\quad s \vee t$	$\vee$I, 7
9	$\quad\quad r$	
10	$\quad\quad t$	$\to$E, 3, 9
11	$\quad\quad s \vee t$	$\vee$I, 10
12	$\quad s \vee t$	$\vee$E, 5, 6–8, 9–11
13	$p \to (s \vee t)$	$\to$E, 4–12

(f) $p \to (q \wedge r), q \vee s \to t, p \vee s \vdash t$.

Solution:

1	$p \to (q \wedge r)$	
2	$q \vee s \to t$	
3	$p \vee s$	
4	$\quad p$	
5	$\quad q \wedge r$	$\to$E, 1, 4
6	$\quad q$	$\wedge$E, 5
7	$\quad q \vee s$	$\vee$I, 6
8	$\quad t$	$\to$E, 2, 7
9	$\quad s$	
10	$\quad q \vee s$	$\vee$E, 9
11	$\quad t$	$\to$E, 2, 10
12	t	$\vee$E, 3, 4–8, 9–11

(g) $(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) \vdash \neg r$.

Solution: Let us take this one step-by-step:

$(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k)$
$\dots$
$\neg r$

First note that we want to prove a negation, therefore the proof follows by contradiction:

$$
\begin{array}{|l|l}
\hline
(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) & \\
\hline
\quad \begin{array}{|l}
\quad r \\
\hline
\quad \ldots \\
\quad \mathbf{F}
\end{array} & \\
\neg r & \neg I
\end{array}
$$

Now notice that our only premise (aside from r) is an implication, so we need to derive its antecedent in order to use it. And in fact, we can deduce $t \to (r \vee p)$ from r:

$$
\begin{array}{|l|l}
\hline
(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) & \\
\hline
\quad \begin{array}{|l}
\quad r \\
\hline
\quad \ldots \\
\quad t \to (r \vee p) \\
\quad (\neg r \vee k) \wedge \neg k \\
\quad \neg r \vee k \\
\quad \neg k \\
\quad \ldots \\
\quad \mathbf{F}
\end{array} & \begin{array}{l}
\\
\\
\\
\to E \\
\wedge E \\
\wedge E \\
\\
\\
\end{array} \\
\neg r & \neg I
\end{array}
$$

Now note that $t \to (r \vee p)$ is an implication therefore we deduce it by assuming t to deduce $r \vee p$.

$$
\begin{array}{|l|l}
\hline
(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k) & \\
\hline
\quad \begin{array}{|l}
\quad r \\
\hline
\quad \begin{array}{|l}
\quad t \\
\hline
\quad r \vee p
\end{array} \\
\quad t \to (r \vee p) \\
\quad (\neg r \vee k) \wedge \neg k \\
\quad \neg r \vee k \\
\quad \neg k \\
\quad \ldots \\
\quad \mathbf{F}
\end{array} & \begin{array}{l}
\\
\\
\vee I \\
\to I \\
\to E \\
\wedge E \\
\wedge E \\
\\
\\
\end{array} \\
\neg r & \neg I
\end{array}
$$

And given that we now have deduced the disjuntion $\neg r \vee k$, then we now deduce $\mathbf{F}$ by elimination

of that disjuntion, thus completing the proof:

1	$(t \to (r \vee p)) \to ((\neg r \vee k) \wedge \neg k)$	
2	$\quad r$	
3	$\quad\quad t$	
4	$\quad\quad r \vee p$	$\vee$I, 2
5	$\quad t \to (r \vee p)$	$\to$I, 3–4
6	$\quad (\neg r \vee k) \wedge \neg k$	$\to$E, 1, 5
7	$\quad \neg r \vee k$	$\wedge$E, 6
8	$\quad \neg k$	$\wedge$E, 6
9	$\quad\quad \neg r$	
10	$\quad\quad \mathbf{F}$	**F**I, 2, 9
11	$\quad\quad k$	
12	$\quad\quad \mathbf{F}$	**F**I, 8, 11
13	$\quad \mathbf{F}$	$\vee$E, 7, 9–10, 11–12
14	$\neg r$	$\neg$I, 2–13

2.5) Show that:

(a) $(\delta \wedge \neg\varphi) \to \psi, \neg\psi, \delta \vdash \varphi$

Solution:

1	$(\delta \wedge \neg\varphi) \to \psi$	
2	$\neg\psi$	
3	δ	
4	$\quad\quad \neg\varphi$	
5	$\quad\quad \delta \wedge \neg\varphi$	$\wedge$I, 3, 4
6	$\quad\quad \psi$	$\to$E, 1, 5
7	$\quad\quad \mathbf{F}$	**F**I, 2, 6
8	$\neg\neg\varphi$	$\neg$I, 4–7
9	φ	$\neg$E, 8

(b) $\vdash (\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$

Solution:

1	$\delta \to \psi$	
2	$\quad \neg\delta \to \neg\varphi$	
3	$\quad\quad \varphi$	
4	$\quad\quad \neg\neg\varphi$	$\neg\neg$I, 3
5	$\quad\quad \neg\neg\delta$	MT, 2, 4
6	$\quad\quad \delta$	$\neg$E, 5
7	$\quad\quad \psi$	$\to$E, 1, 6
8	$\quad \varphi \to \psi$	$\to$I, 3–7
9	$(\neg\delta \to \neg\varphi) \to (\varphi \to \psi)$	$\to$I, 2–8
10	$(\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$	$\to$I, 1–9

Without derived rules:

1	$\delta \to \psi$	
2	$\quad \neg\delta \to \neg\varphi$	
3	$\quad\quad \varphi$	
4	$\quad\quad\quad \neg\psi$	
5	$\quad\quad\quad\quad \delta$	
6	$\quad\quad\quad\quad \psi$	$\to$E, 1, 5
7	$\quad\quad\quad\quad \mathbf{F}$	**F**I, 4, 6
8	$\quad\quad\quad \neg\delta$	$\neg$I, 5–7
9	$\quad\quad\quad \neg\varphi$	$\to$E, 2, 8
10	$\quad\quad\quad \mathbf{F}$	**F**I, 3, 9
11	$\quad\quad \neg\neg\psi$	$\neg$I, 4–10
12	$\quad\quad \psi$	$\neg$E, 11
13	$\quad \varphi \to \psi$	$\to$I, 3–12
14	$(\neg\delta \to \neg\varphi) \to (\varphi \to \psi)$	$\to$I, 2–13
15	$(\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$	$\to$I, 1–14

Using the excluded middle:

1	$\delta \to \psi$	
2	$\quad \neg\delta \to \neg\varphi$	
3	$\quad\quad \varphi$	
4	$\quad\quad \delta \vee \neg\delta$	TE
5	$\quad\quad\quad \delta$	
6	$\quad\quad\quad \psi$	$\to$E, 1, 5
7	$\quad\quad\quad \neg\delta$	
8	$\quad\quad\quad \neg\varphi$	$\to$E, 2, 7
9	$\quad\quad\quad \mathbf{F}$	**F**I, 3, 8
10	$\quad\quad\quad \psi$	**F**E, 9
11	$\quad\quad \psi$	$\vee$E, 4, 5–6, 7–10
12	$\quad \varphi \to \psi$	$\to$I, 3–12
13	$(\neg\delta \to \neg\varphi) \to (\varphi \to \psi)$	$\to$I, 2–13
14	$(\delta \to \psi) \to ((\neg\delta \to \neg\varphi) \to (\varphi \to \psi))$	$\to$I, 1–14

(c) $\vdash (\delta \wedge \varphi) \to \varphi$

Solution:

1	$\quad \delta \wedge \varphi$	
2	$\quad \varphi$	$\wedge$E, 1
3	$(\delta \wedge \varphi) \to \varphi$	$\to$I, 1–2

(d) $\vdash \delta \to (\neg\varphi \to \neg(\delta \to \varphi))$

Solution:

1	$\quad \delta$	
2	$\quad\quad \neg\varphi$	
3	$\quad\quad\quad \delta \to \varphi$	
4	$\quad\quad\quad \varphi$	$\to$E, 1, 3
5	$\quad\quad\quad \mathbf{F}$	**F**I, 2, 4
6	$\quad\quad \neg(\delta \to \varphi)$	$\neg$I, 3–5
7	$\quad \neg\varphi \to \neg(\delta \to \varphi)$	$\to$I, 2–6
8	$\delta \to (\neg\varphi \to \neg(\delta \to \varphi))$	$\to$I, 1–7

(e) $\vdash ((\varphi \to \delta) \wedge (\delta \to \varphi)) \to ((\varphi \vee \delta) \to (\varphi \wedge \delta))$

Solution:

1	$(\varphi \to \delta) \wedge (\delta \to \varphi)$	
2	$\varphi \vee \delta$	
3	φ	
4	$\varphi \to \delta$	$\wedge$E, 1
5	δ	$\to$E, 3, 4
6	$\varphi \wedge \delta$	$\wedge$I, 3, 5
7	δ	
8	$\delta \to \varphi$	$\wedge$E, 1
9	φ	$\to$E, 7, 8
10	$\varphi \wedge \delta$	$\wedge$I, 7, 9
11	$\varphi \wedge \delta$	$\vee$E, 2, 3–6, 7–10
12	$(\varphi \vee \delta) \to (\varphi \wedge \delta)$	$\to$I, 2–11
13	$((\varphi \to \delta) \wedge (\delta \to \varphi)) \to ((\varphi \vee \delta) \to (\varphi \wedge \delta))$	$\to$I, 1–12

2.6) Indicate which of the following deductions are valid and which are not:

(a) $\neg\delta, \delta \vee \varphi \vdash \varphi$

Solution:

1	$\neg\delta$	
2	$\delta \vee \varphi$	
3	δ	
4	**F**	**F**I, 1, 3
5	φ	**F**E, 4
6	φ	
7	φ	R, 6
8	φ	$\vee$E, 2, 3–5, 6–7

(b) $\varphi \wedge \neg\varphi \vdash \neg(\delta \to \psi) \wedge (\delta \to \psi)$

Solution:

1	$\varphi \wedge \neg\varphi$	
2	φ	$\wedge$E, 1
3	$\neg\varphi$	$\wedge$E, 1
4	F	**F**I, 2, 3
5	$\neg(\delta \to \psi) \wedge (\delta \to \psi)$	**F**E, 4

(c) $\neg(\neg\delta \vee \varphi) \vdash \delta$

Solution:

1	$\neg(\neg\delta \vee \varphi)$	
2	$\quad\neg\delta$	
3	$\quad\neg\delta \vee \varphi$	$\vee$I, 2
4	$\quad\mathbf{F}$	**F**I, 1, 3
5	$\neg\neg\delta$	$\neg$I, 2–4
6	δ	$\neg$E, 5

(d) $\delta \vee \psi, \neg\psi \vee \varphi \vdash \delta \vee \varphi$

Solution:

1	$\delta \vee \psi$	
2	$\neg\psi \vee \varphi$	
3	$\quad\delta$	
4	$\quad\delta \vee \varphi$	$\vee$I, 3
5	$\quad\psi$	
6	$\quad\quad\neg\psi$	
7	$\quad\quad\mathbf{F}$	**F**I, 5, 6
8	$\quad\quad\delta \vee \varphi$	**F**E, 7
9	$\quad\quad\varphi$	
10	$\quad\quad\delta \vee \varphi$	$\vee$I, 9
11	$\quad\delta \vee \varphi$	$\vee$E, 2, 6–8, 9–10
12	$\delta \vee \varphi$	$\vee$E, 1, 3–4, 5–11

(e) $\vdash (\neg\delta \to \delta) \to \delta$

Solution:

1	$\quad\neg\delta \to \delta$	
2	$\quad\quad\neg\delta$	
3	$\quad\quad\delta$	$\to$E, 1, 2
4	$\quad\quad\mathbf{F}$	**F**I, 2, 3
5	$\quad\neg\neg\delta$	$\neg$I, 2–4
6	$\quad\delta$	$\neg$E, 5
7	$(\neg\delta \to \delta) \to \delta$	$\to$I, 1–6

(f) $\neg\delta \to \neg\psi \vdash \neg\psi \to \neg\delta$

Solution: $\neg\delta \to \neg\psi \nvdash \neg\psi \to \neg\delta$. We can show the $\neg\delta \to \neg\psi \not\models \neg\psi \to \neg\delta$. Note that if we consider an assignment ν such that $\not\models_\nu \psi$ and $\models_\nu \delta$, then $\models_\nu \neg\delta \to \neg\psi$ and $\not\models_\nu \neg\psi \to \neg\delta$.

2.7) Justify the validity or falsity of each of the following statements with a proof or a counter-example:

(a) if $\vdash \psi$, then $\delta_1, \ldots, \delta_n \vdash \psi$;

Solution: True. If $\vdash \psi$, there is a deduction (a sequence of steps) ending in ψ, from which we can build a deduction with premises $\delta_1, \ldots, \delta_n$.

$$\begin{array}{|l} \Delta \\ \psi \end{array} \quad \Longrightarrow \quad \begin{array}{|l} \delta_1 \\ \vdots \\ \delta_n \\ \hline \Delta \\ \psi \end{array}$$

(b) if $\delta_1, \ldots, \delta_n \vdash \psi$, then $\vdash \psi$;
Solution: False. Note that $p \vdash p$, but $\nvdash p$ since $\not\models p$.

(c) if $\delta_1, \delta_2 \vdash \psi$, then $\delta_2, \delta_1 \vdash \psi$;
Solution: True. If $\delta_1, \delta_2 \vdash \psi$, there is a deduction with premises δ_1, δ_2 ending in ψ, from which we can build a deduction with premises δ_2, δ_1.

$$\begin{array}{|l} \delta_1 \\ \delta_2 \\ \hline \Delta \\ \psi \end{array} \quad \Longrightarrow \quad \begin{array}{|l} \delta_2 \\ \delta_1 \\ \hline \Delta \\ \psi \end{array}$$

(d) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1, \delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;
Solution: True. If $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1, \delta_2 \vdash \varphi$, there deductions with premises δ_1, δ_2 ending in $\varphi \to \psi$ and φ respectively, from which we can build a deduction for ψ with premises δ_2, δ_1.

$$\begin{array}{|l} \delta_1 \\ \delta_2 \\ \hline \Delta_1 \\ \varphi \to \psi \end{array} \qquad \begin{array}{|l} \delta_1 \\ \delta_2 \\ \hline \Delta_2 \\ \psi \end{array} \quad \Longrightarrow \quad \begin{array}{|l} \delta_2 \\ \delta_1 \\ \hline \Delta_1 \\ \varphi \to \psi \\ \Delta_2 \\ \varphi \\ \psi \end{array}$$

(e) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;
Solution: True. If $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then it also follows that $\delta_1, \delta_2 \vdash \varphi$, therefore $\delta_1, \delta_2 \vdash \psi$ by $(\mathtt{d})$.

(f) if $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then $\delta_1 \vdash \psi$;
Solution: False. If $\delta_1 = p$ and $\delta_2 = p \to q$ then $p, p \to q \vdash p \to q$ and $p \vdash p$ but $p \nvdash q$.

(g) if $\delta_1 \vdash \varphi \to \psi$ and $\delta_2 \vdash \varphi$, then $\delta_1, \delta_2 \vdash \psi$;
Solution: True. If $\delta_1 \vdash \varphi \to \psi$ and $\delta_1 \vdash \varphi$, then it also follows that $\delta_1, \delta_2 \vdash \varphi \to \psi$ and $\delta_1, \delta_2 \vdash \varphi$, therefore $\delta_1, \delta_2 \vdash \psi$ by $(\mathtt{d})$.

(h) $\delta, \delta \vdash \psi$ if and only if $\delta \vdash \psi$.
Solution: Note that the relation $\vdash$ is defined between a set of premises and a formula. And $\{\delta, \delta\} = \{\delta\}$, therefore the equivalence is true.

2.8) Show, without using the completeness of DN, the following deductive equivalences. Note that you have to show that the second formula is deduced from the first formula and vice versa.

(a) $\neg\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

Solution:

1	$\neg\delta_1 \wedge \delta_2$	
2	$\quad\neg(\delta_1 \vee \delta_2)$	
3	$\quad\quad\delta_2$	
4	$\quad\quad\delta_1 \vee \delta_2$	$\vee$I, 3
5	$\quad\quad\mathbf{F}$	**F**I, 2, 4
6	$\quad\neg\delta_2$	$\neg$I, 3–5
7	$\quad\delta_2$	$\wedge$E, 1
8	$\quad\mathbf{F}$	**F**I, 6, 7
9	$\neg\neg(\delta_1 \vee \delta_2)$	$\neg$I, 2–8
10	$\delta_1 \vee \delta_2$	$\neg$E, 9

Using the excluded middle:

1	$\neg\delta_1 \wedge \delta_2$	
2	$\delta_1 \vee \neg\delta_1$	TE
3	$\quad\delta_1$	
4	$\quad\delta_1 \vee \delta_2$	$\vee$I, 3
5	$\quad\neg\delta_1$	
6	$\quad\delta_2$	$\wedge$E, 1
7	$\quad\delta_1 \vee \delta_2$	$\vee$I, 6
8	$\delta_1 \vee \delta_2$	$\vee$E, 2, 3–4, 5–7

And yet another solution (much simpler!)

1	$\neg\delta_1 \wedge \delta_2$	
2	δ_2	$\wedge$E, 1
3	$\delta_1 \vee \delta_2$	$\vee$I, 2

(b) $\delta_1 \wedge \delta_2 \vdash \delta_1 \vee \delta_2$

Solution:

1	$\delta_1 \wedge \delta_2$	
2	δ_1	$\wedge$E, 1
3	$\delta_1 \vee \delta_2$	$\vee$I, 2

(c) $(\psi \wedge \varphi) \wedge \delta \dashv\vdash \psi \wedge (\varphi \wedge \delta)$

Solution:

1	$\psi \wedge (\varphi \wedge \delta)$	
2	ψ	$\wedge$E, 1
3	$\varphi \wedge \delta$	$\wedge$E, 1
4	φ	$\wedge$E, 3
5	$\psi \wedge \varphi$	$\wedge$I, 2, 4
6	δ	$\wedge$E, 3
7	$(\psi \wedge \varphi) \wedge \delta$	$\wedge$I, 5, 6

1	$(\psi \wedge \varphi) \wedge \delta$	
2	$\psi \wedge \varphi$	$\wedge$E, 1
3	ψ	$\wedge$E, 2
4	φ	$\wedge$E, 2
5	δ	$\wedge$E, 1
6	$\varphi \wedge \delta$	$\wedge$I, 4, 5
7	$\psi \wedge (\varphi \wedge \delta)$	$\wedge$I, 3, 6

(d) $\neg(\neg\delta_1 \wedge \neg\delta_2) \dashv\vdash \delta_1 \vee \delta_2$

Solution:

1	$\neg(\neg\delta_1 \wedge \neg\delta_2)$	
2	$\quad\neg(\delta_1 \vee \delta_2)$	
3	$\quad\quad\delta_1$	
4	$\quad\quad\delta_1 \vee \delta_2$	$\vee$I, 3
5	$\quad\quad$**F**	**F**I, 2, 4
6	$\quad\neg\delta_1$	$\neg$I, 3–5
7	$\quad\quad\delta_2$	
8	$\quad\quad\delta_1 \vee \delta_2$	$\vee$I, 7
9	$\quad\quad$**F**	**F**I, 2, 8
10	$\quad\neg\delta_2$	$\neg$I, 7–9
11	$\quad\neg\delta_1 \wedge \neg\delta_2$	$\wedge$I, 6, 10
12	$\quad$**F**	**F**I, 1, 11
13	$\neg\neg(\delta_1 \vee \delta_2)$	$\neg$I, 2–12
14	$\delta_1 \vee \delta_2$	$\neg$E, 13

1	$\delta_1 \vee \delta_2$	
2	$\quad\neg\delta_1 \wedge \neg\delta_2$	
3	$\quad\quad\delta_1$	
4	$\quad\quad\neg\delta_1$	$\wedge$E, 2
5	$\quad\quad$**F**	**F**I, 3, 4
6	$\quad\quad\delta_2$	
7	$\quad\quad\neg\delta_2$	$\wedge$E, 2
8	$\quad\quad$**F**	**F**I, 6, 7
9	$\quad$**F**	$\vee$E, 1, 3–5, 6–8
10	$\neg(\neg\delta_1 \wedge \neg\delta_2)$	$\neg$I, 2–9

Let us look at the proof on the left step-by-step:

$$\begin{array}{|l} \neg(\neg\delta_1 \wedge \neg\delta_2) \\ \hline \quad\vdots \\ \quad\delta_1 \vee \delta_2 \end{array}$$

Note that the only premise we have is a negation that can only be used to introduce **F**, which

means we need more premises. Let us then follow by contradiction:

$\neg(\neg\delta_1 \wedge \neg\delta_2)$	
$\quad\neg(\delta_1 \vee \delta_2)$	
$\quad\quad\ldots$	
$\quad\quad\mathbf{F}$	
$\neg\neg(\delta_1 \vee \delta_2)$	$\neg$I
$\delta_1 \vee \delta_2$	$\neg$E

Now to deduce $\mathbf{F}$, we can either try to deduce $\delta_1 \vee \delta_2$ (which can be obtained from either δ_1 or δ_2) or $\neg\delta_1 \wedge \neg\delta_2$ (which can be obtained from $\neg\delta_1$ and $\neg\delta_2$). But neither δ_1 or δ_2 are consequences of the premises whereas both $\neg\delta_1$ and $\neg\delta_2$ are, since both are negations.

$\neg(\neg\delta_1 \wedge \neg\delta_2)$	
$\quad\neg(\delta_1 \vee \delta_2)$	
$\quad\quad\ldots$	
$\quad\quad\neg\delta_1$	
$\quad\quad\ldots$	
$\quad\quad\neg\delta_2$	
$\quad\quad\neg\delta_1 \wedge \neg\delta_2$	
$\quad\quad\mathbf{F}$	
$\neg\neg(\delta_1 \vee \delta_2)$	$\neg$I
$\delta_1 \vee \delta_2$	$\neg$E

This leads to two sub-deductions by contradiction, which can easily be obtained, thus completing the deduction:

1	$\neg(\neg\delta_1 \wedge \neg\delta_2)$	
2	$\quad\neg(\delta_1 \vee \delta_2)$	
3	$\quad\quad\delta_1$	
4	$\quad\quad\delta_1 \vee \delta_2$	$\vee$I, 3
5	$\quad\quad\mathbf{F}$	**F**I, 2, 4
6	$\quad\neg\delta_1$	$\neg$I, 3–5
7	$\quad\quad\delta_2$	
8	$\quad\quad\delta_1 \vee \delta_2$	$\vee$I, 7
9	$\quad\quad\mathbf{F}$	**F**I, 2, 8
10	$\quad\neg\delta_2$	$\neg$I, 7–9
11	$\quad\neg\delta_1 \wedge \neg\delta_2$	$\wedge$I, 6, 10
12	$\quad\mathbf{F}$	**F**I, 1, 11
13	$\neg\neg(\delta_1 \vee \delta_2)$	$\neg$I, 2–12
14	$\delta_1 \vee \delta_2$	$\neg$E, 13

(e) $\delta_1 \to \delta_2 \dashv\vdash \neg(\delta_1 \wedge \neg\delta_2)$

Solution:

1	$\delta_1 \to \delta_2$	
2	| $\delta_1 \wedge \neg\delta_2$	
3	| δ_1	$\wedge$E, 2
4	| δ_2	$\to$E, 1, 3
5	| $\neg\delta_2$	$\wedge$E, 2
6	| **F**	**F**I, 4, 5
7	$\neg(\delta_1 \wedge \neg\delta_2)$	$\neg$I, 2–6

1	$\neg(\delta_1 \wedge \neg\delta_2)$	
2	| δ_1	
3	| | $\neg\delta_2$	
4	| | $\delta_1 \wedge \neg\delta_2$	$\wedge$I, 2, 3
5	| | **F**	**F**I, 1, 4
6	| $\neg\neg\delta_2$	$\neg$I, 3–5
7	| δ_2	$\neg$E, 6
8	$\delta_1 \to \delta_2$	$\to$I, 2–7

(f) $\delta_1 \wedge \delta_2 \dashv\vdash \neg(\neg\delta_1 \vee \neg\delta_2)$

1	$\delta_1 \wedge \delta_2$	
2	| $\neg\delta_1 \vee \neg\delta_2$	
3	| | $\neg\delta_1$	
4	| | δ_1	$\wedge$E, 1
5	| | **F**	**F**I, 3, 4
6	| | $\neg\delta_2$	
7	| | δ_2	$\wedge$E, 1
8	| | **F**	**F**I, 6, 7
9	| **F**	$\vee$E, 2, 3–5, 6–8
10	$\neg(\neg\delta_1 \vee \neg\delta_2)$	$\neg$I, 2–9

1	$\neg(\neg\delta_1 \vee \neg\delta_2)$	
2	| $\neg\delta_1$	
3	| $\neg\delta_1 \vee \neg\delta_2$	$\vee$I, 3
4	| **F**	**F**I, 1, 3
5	$\neg\neg\delta_1$	$\neg$I, 2–4
6	δ_1	$\neg$E, 5
7	| $\neg\delta_2$	
8	| $\neg(\neg\delta_1 \vee \neg\delta_2)$	$\vee$I, 7
9	| **F**	**F**I, 1, 8
10	$\neg\neg\delta_2$	$\neg$I, 7–9
11	δ_2	$\neg$E, 10
12	$\delta_1 \wedge \delta_2$	$\wedge$I, 6, 11

Let us look at the proof on the right step-by-step:

$\neg(\neg\delta_1 \vee \neg\delta_2)$
...
$\delta_1 \wedge \delta_2$

Because our conclusion is a conjunction, then one needs to derive both components:

$\neg(\neg\delta_1 \vee \neg\delta_2)$	
$\ldots$	
δ_1	
$\ldots$	
δ_2	
$\delta_1 \wedge \delta_2$	$\wedge$I

Now, because the only premise we have available is a negation, both δ_1 and δ_2 will be deduced by contradiction:

$\neg(\neg\delta_1 \vee \neg\delta_2)$	
$\quad\neg\delta_1$	
$\quad\ldots$	
$\quad\mathbf{F}$	
$\neg\neg\delta_1$	$\neg$I
δ_1	$\neg$E
$\quad\neg\delta_2$	
$\quad\ldots$	
$\quad\mathbf{F}$	
$\neg\neg\delta_2$	$\neg$I
δ_2	$\neg$E
$\delta_1 \wedge \delta_2$	$\wedge$I

These are easily obtained using premise $\neg(\neg\delta_1 \vee \neg\delta_2)$, thus concluding the derivation:

1	$\neg(\neg\delta_1 \vee \neg\delta_2)$	
2	$\quad\neg\delta_1$	
3	$\quad\neg\delta_1 \vee \neg\delta_2$	$\vee$I, 3
4	$\quad\mathbf{F}$	**F**I, 1, 3
5	$\neg\neg\delta_1$	$\neg$I, 2–4
6	δ_1	$\neg$E, 5
7	$\quad\neg\delta_2$	
8	$\quad\neg(\neg\delta_1 \vee \neg\delta_2)$	$\vee$I, 7
9	$\quad\mathbf{F}$	**F**I, 1, 8
10	$\neg\neg\delta_2$	$\neg$I, 7–9
11	δ_2	$\neg$E, 10
12	$\delta_1 \wedge \delta_2$	$\wedge$I, 6, 11

(g) $(\varphi \vee \psi) \to \delta \dashv\vdash (\varphi \to \delta) \wedge (\psi \to \delta)$

Solution:

1	$(\varphi \vee \psi) \to \delta$	
2	$\quad \varphi$	
3	$\quad \varphi \vee \psi$	$\vee$I, 2
4	$\quad \delta$	$\to$E, 1, 3
5	$\varphi \to \delta$	$\to$I, 2–4
6	$\quad \psi$	
7	$\quad \varphi \vee \psi$	$\vee$I, 6
8	$\quad \delta$	$\to$E, 1, 7
9	$\psi \to \delta$	$\to$I, 6–8
10	$(\varphi \to \delta) \wedge (\psi \to \delta)$	$\wedge$I, 5, 9

1	$(\varphi \to \delta) \wedge (\psi \to \delta)$	
2	$\quad \varphi \vee \psi$	
3	$\quad\quad \varphi$	
4	$\quad\quad \varphi \to \delta$	$\wedge$E, 1
5	$\quad\quad \delta$	$\to$E, 3, 4
6	$\quad\quad \psi$	
7	$\quad\quad \psi \to \delta$	$\wedge$E, 1
8	$\quad\quad \delta$	$\to$E, 6, 7
9	$\quad \delta$	$\vee$E, 2, 3–5, 6–8
10	$(\varphi \vee \psi) \to \delta$	$\to$I, 2–9

(h) $(\delta \vee \varphi) \wedge (\delta \vee \psi) \dashv\vdash \delta \vee (\varphi \wedge \psi)$

Solution:

1	$(\delta \vee \varphi) \wedge (\delta \vee \psi)$	
2	$\delta \vee \varphi$	$\wedge$E, 1
3	$\delta \vee \psi$	$\wedge$E, 1
4	$\quad \neg(\delta \vee (\varphi \wedge \psi))$	
5	$\quad\quad \delta$	
6	$\quad\quad \delta \vee (\varphi \wedge \psi)$	$\vee$I, 5
7	$\quad\quad \mathbf{F}$	**F**I, 4, 6
8	$\quad\quad \varphi$	
9	$\quad\quad\quad \delta$	
10	$\quad\quad\quad \delta \vee (\varphi \wedge \psi)$	$\vee$I, 9
11	$\quad\quad\quad \mathbf{F}$	**F**I, 4, 10
12	$\quad\quad\quad \psi$	
13	$\quad\quad\quad \varphi \wedge \psi$	$\wedge$I, 8, 12
14	$\quad\quad\quad \delta \vee (\varphi \wedge \psi)$	$\vee$I, 13
15	$\quad\quad\quad \mathbf{F}$	**F**I, 4, 14
16	$\quad\quad \mathbf{F}$	$\vee$E, 3, 9–11, 12–15
17	$\quad \mathbf{F}$	$\vee$E, 2, 5–7, 8–16
18	$\neg\neg(\delta \vee (\varphi \wedge \psi))$	$\neg$I, 4–17
19	$\delta \vee (\varphi \wedge \psi)$	$\neg$E, 18

1	$\delta \vee (\varphi \wedge \psi)$	
2	$\quad \delta$	
3	$\quad \delta \vee \varphi$	$\vee$I, 2
4	$\quad \delta \vee \psi$	$\vee$I, 2
5	$\quad (\delta \vee \varphi) \wedge (\delta \vee \psi)$	$\wedge$I, 3, 4
6	$\quad \varphi \wedge \psi$	
7	$\quad \varphi$	$\wedge$E, 6
8	$\quad \delta \vee \varphi$	$\vee$I, 7
9	$\quad \psi$	$\wedge$E, 6
10	$\quad \delta \vee \varphi$	$\vee$I, 9
11	$\quad (\delta \vee \varphi) \wedge (\delta \vee \psi)$	$\wedge$I, 8, 10
12	$(\delta \vee \varphi) \wedge (\delta \vee \psi)$	$\vee$E, 1, 2–5, 6–11

Let us look at the proof on the left step-by-step:

$$\begin{array}{|l}
\hline
(\delta \vee \varphi) \wedge (\delta \vee \psi) \\
\hline
\cdots \\
\delta \vee (\varphi \wedge \psi)
\end{array}$$

Notice that our only premise is a conjunction therefore we need to use elimination to use its components:

$$\begin{array}{|ll}
\hline
(\delta \vee \varphi) \wedge (\delta \vee \psi) & \\
\hline
\delta \vee \varphi & \wedge\mathrm{E} \\
\delta \vee \psi & \wedge\mathrm{E} \\
\cdots & \\
\delta \vee (\varphi \wedge \psi) &
\end{array}$$

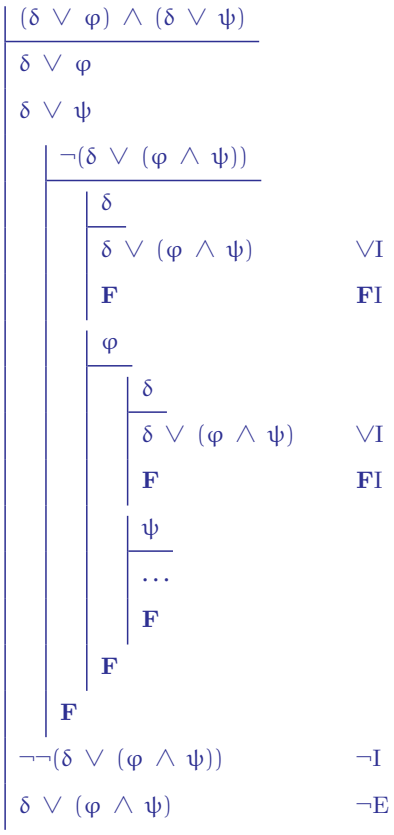
Now the conclusion is a disjunction, by neither δ nor $\varphi \wedge \psi$ can be deduced from the premises. So we need more premises, which leads to a proof by contradiction:

$$\begin{array}{|ll}
\hline
(\delta \vee \varphi) \wedge (\delta \vee \psi) & \\
\hline
\delta \vee \varphi & \wedge\mathrm{E} \\
\delta \vee \psi & \wedge\mathrm{E} \\
\quad \begin{array}{|l} \hline \neg(\delta \vee (\varphi \wedge \psi)) \\ \hline \cdots \\ \mathbf{F} \end{array} & \\
\neg\neg(\delta \vee (\varphi \wedge \psi)) & \neg\mathrm{I} \\
\delta \vee (\varphi \wedge \psi) & \neg\mathrm{E}
\end{array}$$

Now we need to use our disjuntions to conclude $\mathbf{F}$, let us start by eliminating the first one. Note that from δ and $\neg(\delta \vee (\varphi \wedge \psi))$ one can easily deduce $\mathbf{F}$.

$$\begin{array}{|ll}
\hline
(\delta \vee \varphi) \wedge (\delta \vee \psi) & \\
\hline
\delta \vee \varphi & \\
\delta \vee \psi & \\
\quad \begin{array}{|ll} \hline \neg(\delta \vee (\varphi \wedge \psi)) & \\ \hline \quad \begin{array}{|l} \hline \delta \\ \hline \delta \vee (\varphi \wedge \psi) \\ \mathbf{F} \end{array} & \vee\mathrm{I} \\ & \mathbf{F}\mathrm{I} \\ \quad \begin{array}{|l} \hline \varphi \\ \hline \cdots \\ \mathbf{F} \end{array} & \\ \mathbf{F} & \end{array} & \\
\neg\neg(\delta \vee (\varphi \wedge \psi)) & \neg\mathrm{I} \\
\delta \vee (\varphi \wedge \psi) & \neg\mathrm{E}
\end{array}$$

Now, for the second sub-deduction we will eliminate the second deduction:



Now notice that in this sub-deduction we have φ and ψ, from which we can get $\delta \vee (\varphi \wedge \psi)$

and deduce **F**, which concludes the deduction.

1	$(\delta \vee \varphi) \wedge (\delta \vee \psi)$	
2	$\delta \vee \varphi$	$\wedge$E, 1
3	$\delta \vee \psi$	$\wedge$E, 1
4	$\neg(\delta \vee (\varphi \wedge \psi))$	
5	δ	
6	$\delta \vee (\varphi \wedge \psi)$	$\vee$I, 5
7	**F**	**F**I, 4, 6
8	φ	
9	δ	
10	$\delta \vee (\varphi \wedge \psi)$	$\vee$I, 9
11	**F**	**F**I, 4, 10
12	ψ	
13	$\varphi \wedge \psi$	$\wedge$I, 8, 12
14	$\delta \vee (\varphi \wedge \psi)$	$\vee$I, 13
15	**F**	**F**I, 4, 14
16	**F**	$\vee$E, 3, 9–11, 12–15
17	**F**	$\vee$E, 2, 5–7, 8–16
18	$\neg\neg(\delta \vee (\varphi \wedge \psi))$	$\neg$I, 4–17
19	$\delta \vee (\varphi \wedge \psi)$	$\neg$E, 18

(i) $\varphi \to (\psi \wedge \delta) \dashv\vdash (\varphi \to \psi) \wedge (\varphi \to \delta)$

Solution:

1	$\varphi \to (\psi \wedge \delta)$	
2	φ	
3	$\psi \wedge \delta$	$\to$E, 1, 2
4	ψ	$\wedge$E, 3
5	$\varphi \to \psi$	$\to$I, 2–4
6	φ	
7	$\psi \wedge \delta$	$\to$E, 1, 6
8	δ	$\wedge$E, 7
9	$\varphi \to \delta$	$\to$I, 6–8
10	$(\varphi \to \psi) \wedge (\varphi \to \delta)$	$\wedge$I, 5, 9

1	$(\varphi \to \psi) \wedge (\varphi \to \delta)$	
2	φ	
3	$\varphi \to \psi$	$\wedge$E, 1
4	ψ	$\to$E, 2, 3
5	$\varphi \to \delta$	$\wedge$E, 1
6	δ	$\to$E, 2, 5
7	$\psi \wedge \delta$	$\wedge$I, 5, 7
8	$\varphi \to (\psi \wedge \delta)$	$\to$I, 2–7

(j) $\delta_1 \wedge \neg\delta_2 \vdash \neg(\delta_1 \to \delta_2)$

Solution:

1	$\delta_1 \wedge \neg\delta_2$	
2	$\quad \delta_1 \rightarrow \delta_2$	
3	$\quad \delta_1$	$\wedge$E, 1
4	$\quad \delta_2$	$\rightarrow$E, 2, 3
5	$\quad \neg\delta_2$	$\wedge$E, 1
6	$\quad \mathbf{F}$	**F**I, 4, 5
7	$\neg(\delta_1 \rightarrow \delta_2)$	$\neg$I, 2–6

2.9) Let Σ and Γ be sets of propositional logic formulas such that for any formula δ one has $\Sigma \vdash \delta$ or $\Gamma \vdash \delta$.

(a) Show that $\Sigma \cup \Gamma$ is not satisfiable.

Solution: Let Σ and Γ be any set of formulas. Since for any formula δ one has $\Sigma \vdash \delta$ or $\Gamma \vdash \delta$, then this is also true for $\delta = \mathbf{F}$, which means that either $\Sigma \vdash \mathbf{F}$ or $\Gamma \vdash \mathbf{F}$, which implies either $\Sigma \vDash \mathbf{F}$ or $\Gamma \vDash \mathbf{F}$. But if $\Sigma \vDash \mathbf{F}$ then Σ is not satisfiable, which implies $\Sigma \cup \Gamma$ is not satisfiable. And if $\Gamma \vDash \mathbf{F}$ then Γ is not satisfiable, which implies $\Sigma \cup \Gamma$ is not satisfiable.

(b) Show that one of the sets Σ or Γ is not satisfiable.

Solution: Again, consider $\delta = \mathbf{F}$, which implies either $\Sigma \vDash \mathbf{F}$ or $\Gamma \vDash \mathbf{F}$. But if $\Sigma \vDash \mathbf{F}$ then Σ is not satisfiable . And if $\Gamma \vDash \mathbf{F}$ then Γ is not satisfiable.